

WEEK – 3 : Write a Program Using JavaScript (Arrays, Functions)

Aim: To design and develop a Contact List application using JavaScript arrays and functions, which allows the user to add, display, search, and delete contacts dynamically.

Description: JavaScript is a scripting language used to make web pages interactive and dynamic. In this experiment, a Contact List application is developed using HTML, CSS, and JavaScript.

An array is used to store multiple contact objects containing the name and phone number of each contact. JavaScript functions are used to perform different operations such as adding a new contact, displaying contacts, deleting a contact, and searching for a contact.

The application provides input fields for entering a contact's name and phone number. When the Add Contact button is clicked, the contact is stored in the array and displayed on the webpage. Each contact has a Delete button that removes the contact from the array. A search box allows the user to filter contacts by name while typing. The total number of contacts is also displayed dynamically.

The program also demonstrates the use of DOM manipulation, `forEach()`, `push()`, `splice()`, template literals, and the `addEventListener()` method.

Concepts Used

Concept	Purpose
Array	Stores multiple contacts
Object	Stores the name and phone number of a contact
Function	Performs specific operations
<code>push()</code>	Adds a contact to the array
<code>splice()</code>	Deletes a contact from the array
<code>forEach()</code>	Iterates through the contacts
DOM Manipulation	Dynamically updates webpage content
<code>getElementById()</code>	Accesses HTML elements
<code>innerHTML</code>	Displays dynamic HTML content
<code>addEventListener()</code>	Detects user actions
<code>includes()</code>	Searches for a contact name
<code>toLowerCase()</code>	Makes searching case-insensitive
Template Literals	Creates dynamic HTML content

Code

HTML Code

```
<!DOCTYPE html>

<html>

<head>
  <title>Contact List</title>
  <link rel="stylesheet" href="./style.css">
</head>

<body>

  <h1>Contact List</h1>

  <div class="info">

    <input
      type="text"
      id="name"
      placeholder="Enter contact name"
    >

    <input
      type="text"
      id="phone"
      placeholder="Enter phone number"
    >

    <button onclick="addContact()">Add Contact</button>

  </div>

  <input
    type="search"
    placeholder="Search the contact"
    id="search"
  >
```

```
<div id="contactList"></div>

<p id="totalContacts">Total Contacts: 0</p>

<script src="./script.js"></script>

</body>

</html>
```

CSS Code

```
body {
    background-color: white;
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 20px;
}

/* Heading */
h1 {
    background-color: cadetblue;
    color: aliceblue;
    width: 250px;
    margin: 20px auto;
    padding: 10px;
    text-align: center;
    border-radius: 12px;
    box-shadow: 5px 5px 15px rgba(0, 0, 0, 0.15);
}

/* Add Contact Section */
.info {
    background-color: rgb(211, 96, 96);
    margin: 20px auto;
    padding: 20px;
```

```
    border-radius: 12px;
    width: 60%;
    box-shadow: 0 12px 40px rgba(0, 0, 0, 0.15);
}
```

```
/* Input boxes */
```

```
input {
    display: block;
    width: 90%;
    margin: 10px auto;
    padding: 12px;
    border: none;
    border-radius: 8px;
    box-sizing: border-box;
    outline: none;
}
```

```
/* Buttons */
```

```
button {
    border: none;
    padding: 10px 20px;
    background-color: cadetblue;
    color: aliceblue;
    border-radius: 8px;
    display: block;
    margin: 15px auto;
    cursor: pointer;
}
```

```
button:hover {
    background-color: rgb(61, 130, 132);
}
```

```
/* Search */
```

```
#search {
    display: block;
```

```
width: 50%;
margin: 20px auto;
padding: 12px;
border: none;
background-color: rgb(211, 96, 96);
color: white;
border-radius: 12px;
box-shadow: 5px 5px 15px rgba(0, 0, 0, 0.15);
outline: none;
}
```

```
#search::placeholder {
    color: bisque;
}
```

```
/* Contact List */
#contactList {
    width: 60%;
    margin: 20px auto;
}
```

```
/* Contact Cards */
#contactList .info {
    background-color: rgb(211, 96, 96);
    color: bisque;
    width: auto;
    margin: 15px 0;
    padding: 20px;
    border-radius: 12px;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}
```

```
/* Contact name */
.info h2 {
    margin-top: 0;
    text-align: center;
}
```

```
}

/* Contact phone */
.info p {
    text-align: center;
}

/* Total contacts */
#totalContacts {
    text-align: center;
    font-size: 18px;
    font-weight: bold;
    margin-top: 25px;
}
```

JavaScript Code

```
let contacts = [];
// Add Contact
function addContact() {

    let name = document.getElementById("name").value;
    let phone = document.getElementById("phone").value;

    if (name === "" || phone === "") {
        alert("Please enter name and phone number");
        return;
    }

    let contact = {
        name: name,
        phone: phone
    };

    contacts.push(contact);
```

```

document.getElementById("name").value = "";
document.getElementById("phone").value = "";

displayContacts();
}

// Display Contacts
function displayContacts() {

    let contactList = document.getElementById("contactList");

    contactList.innerHTML = "";

    contacts.forEach(function(contact, index) {

        contactList.innerHTML += `
            <div class="info">

                <h2>${contact.name}</h2>

                <p>${contact.phone}</p>

                <button onclick="deleteContact(${index})">
                    Delete
                </button>

            </div>
        `;
    });

    document.getElementById("totalContacts").innerText =
        "Total Contacts: " + contacts.length;
}

```

```
// Delete Contact
function deleteContact(index) {

    contacts.splice(index, 1);

    displayContacts();
}

// Search Contact
function searchContact() {

    let searchValue = document
        .getElementById("search")
        .value
        .toLowerCase();

    let contactList = document.getElementById("contactList");

    contactList.innerHTML = "";

    contacts.forEach(function(contact, index) {

        if (contact.name.toLowerCase().includes(searchValue)) {

            contactList.innerHTML += `
                <div class="info">

                    <h2>${contact.name}</h2>

                    <p>${contact.phone}</p>

                    <button onclick="deleteContact(${index})">
                        Delete
                    </button>
                </div>
            `;
        }
    });
}
```



```

        </div>
    `;
    }
});
}

// Search while typing
document
    .getElementById("search")
    .addEventListener("input", searchContact);

```

Output :

The screenshot displays a web interface for managing contacts. It features a header 'Contact List', a form to add new contacts, a search bar, and a list of existing contacts. The contact 'vani' is currently listed.

Name	Phone Number	Action
vani	981247293787	Delete

Total Contacts: 1